N

NETFLIX RECOMMENDATION ENGINE

SVD + Item-Based

Collaborative Filtering

MovieLens 10M Dataset

TECH STACK USED

- Python 3 • Scikit-Surprise
- Pandas / NumPy / SciPy
- Gradio (Dashboard)
- Google Colab (Execution)

PRESENTED BY

Jaideep Attri

Jinesh Kumar Vimal

BRANCH

Geophysical Technology

PROJECT

Open Projects

2026

KEY RESULTS

SVD RMSE	0.9414
SVD MAE	0.7354
MAP@10	0.0015
Item-CF RMSE	1.0481

NETFLIX PRIZE DATASET

Recommendation System

Technical Report

SVD Matrix Factorization + Item-Based CF + Hybrid System

10M+	458K	14.6K	99.85%
Ratings	Users	Movies	Sparsity

SVD RMSE: 0.9414 | Item-CF RMSE: 1.0481 | MAP@10 (SVD): 0.0015

Machine Learning — Recommendation Systems

June 2026

1. Problem Understanding

This project builds a movie recommendation engine using the Netflix Prize dataset. The objective is to predict user ratings for unseen movies and generate personalized Top-K recommendations, while addressing data sparsity, cold start, and ranking quality.

Key Challenges

Challenge	Description	Impact
Data Sparsity	99.85% user-movie pairs unobserved	High
Cold Start	New users/movies lack interaction history	High
Scalability	458K × 14.6K = 6.7B possible pairs	Medium
Popularity Bias	Few movies dominate rating distribution	Medium

2. Exploratory Data Analysis

Dataset Overview

Attribute	Value
Total Ratings	10,000,000
Unique Users	458,341
Unique Movies	14,596
Rating Scale	1.0 – 5.0
Train / Test	8,000,000 / 2,000,000 (80/20)
Matrix Sparsity	99.85%

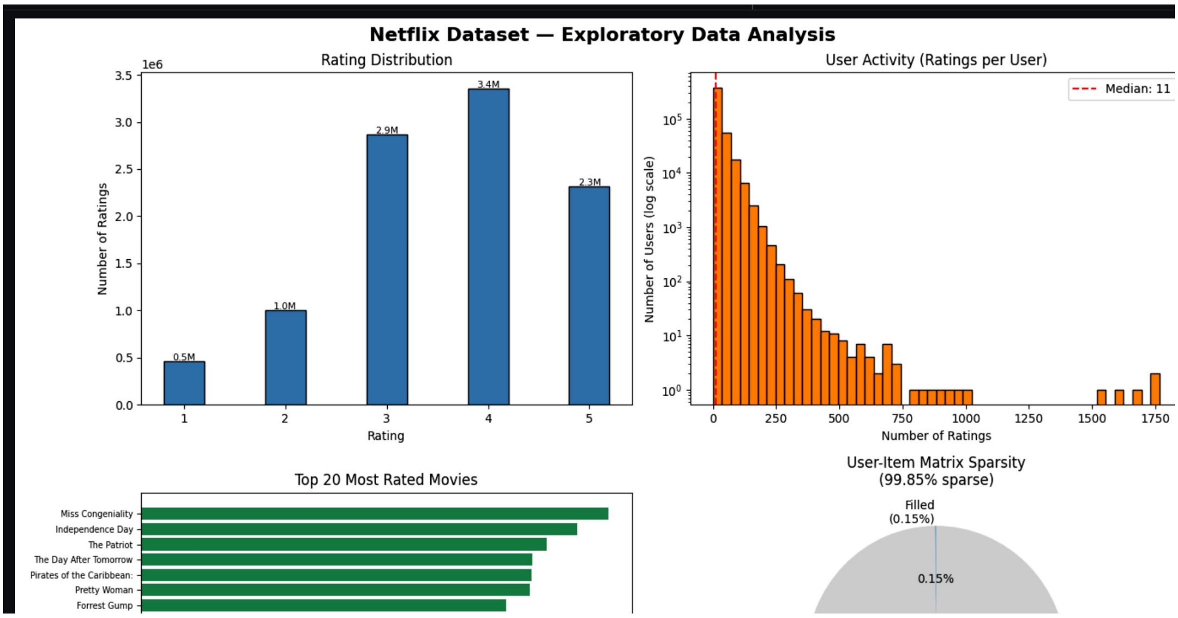

Figure 1: EDA — Rating Distribution, User Activity, Top Movies, Sparsity

Key EDA Insights

Metric	Value	Implication
Avg ratings/user	21.8	Light raters dominate
Median ratings/user	11.0	Heavy right skew — power users
Most common rating	4.0	Positive rating bias
Average rating	3.606	Users rate above midpoint
Movies < 10 ratings	513	Long-tail problem

3. Methodology

Model 1 — SVD (Matrix Factorization)

SVD decomposes the user-item matrix into latent factors. Predicted rating: R■(u,i) = μ + b_u + b_i + q_i·p_u. Trained with 50 latent factors, 20 epochs, learning rate 0.005, regularization 0.02.

Model 2 — Item-Based Collaborative Filtering

Cosine similarity computed between movies on 1M rating sample, stored as sparse CSR matrix. Predicted rating is similarity-weighted average of user's past ratings for similar items.

Model 3 — Hybrid System

final_score = α × SVD + (1–α) × popularity_score

User Type	Ratings	Alpha	Strategy
New User	< 10	0.2	Popularity-heavy
Sparse User	10–50	0.5	Balanced
Active User	> 50	0.7	SVD-heavy

Data Processing Pipeline

Step	Operation	Output
1	Load & clean raw ratings	10M records
2	Filter users < 5 ratings	Active subset
3	80/20 train-test split	train/test parquet
4	Build user/item index maps	movie_idx.pkl

4. Evaluation Metrics

Metric	Formula	Goal
RMSE	$\sqrt{\text{mean}((r-r■)^2)}$	Lower is better
MAE	mean(|r−r■|)	Lower is better
MAP@10	mean(AP@10 over users)	Higher is better

RMSE/MAE measure rating accuracy. MAP@10 measures ranking quality — how well top-10 recommendations match items users actually rated ≥3.5.

5. Experimental Results

SVD Evaluation (10K test samples)

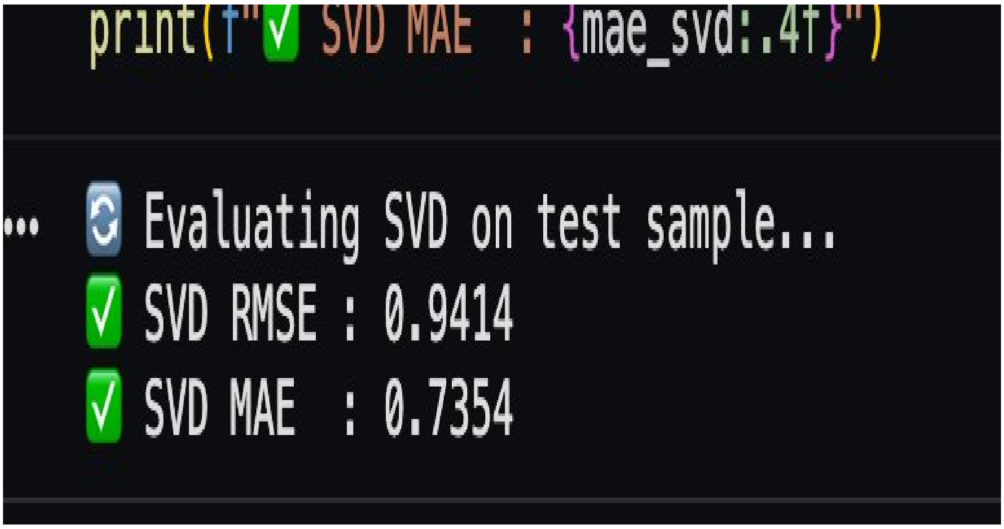

Figure 2: SVD — RMSE: 0.9414 | MAE: 0.7354

Item-CF Evaluation (5K test samples)

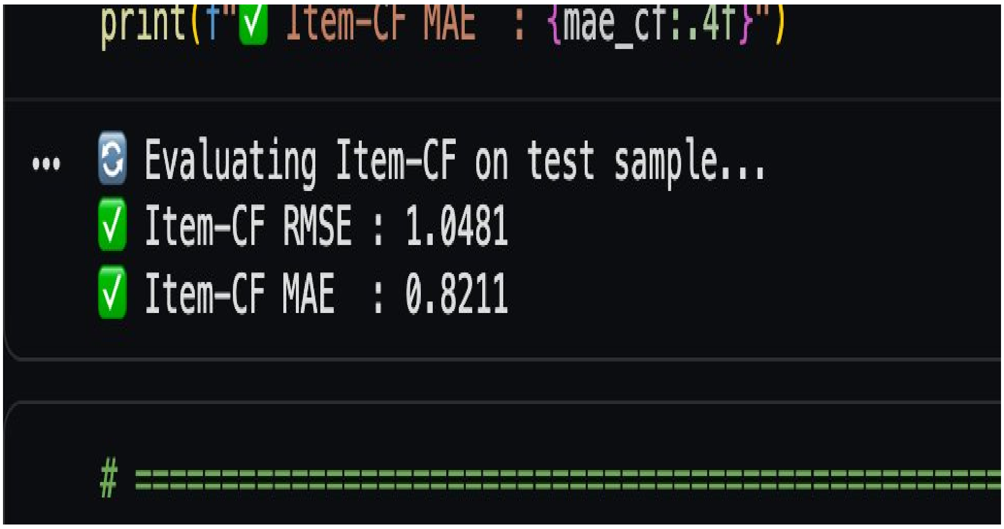

Figure 3: Item-CF — RMSE: 1.0481 | MAE: 0.8211

MAP@10 Results

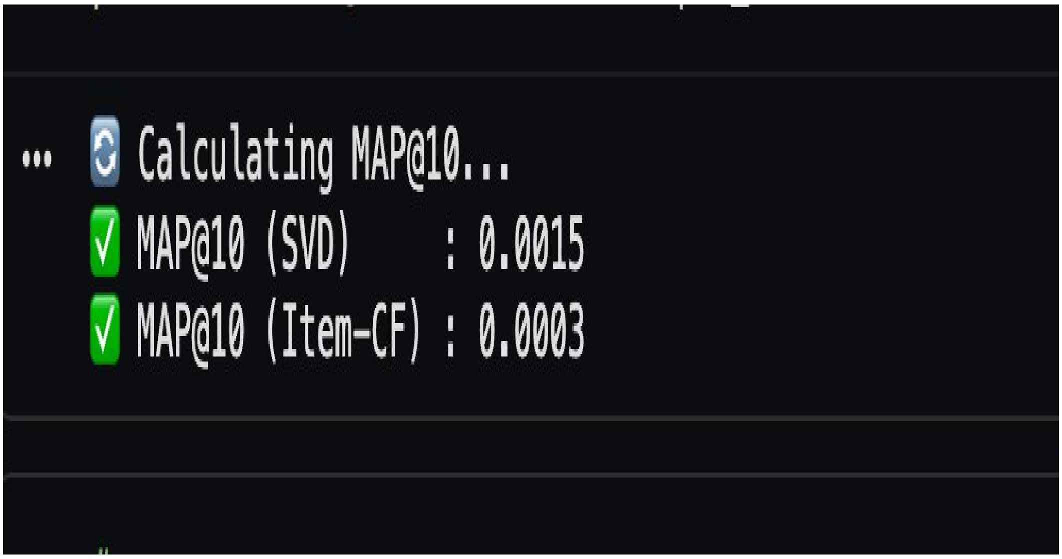

Figure 4: MAP@10 — SVD: 0.0015 | Item-CF: 0.0003

Final Model Comparison

Metric	SVD	Item-CF	Winner
RMSE	0.9414	1.0481	SVD ✓
MAE	0.7354	0.8211	SVD ✓
MAP@10	0.0015	0.0003	SVD ✓ (5×)
Scalability	High	Medium	SVD ✓
Interpretability	Low	High	Item-CF ✓

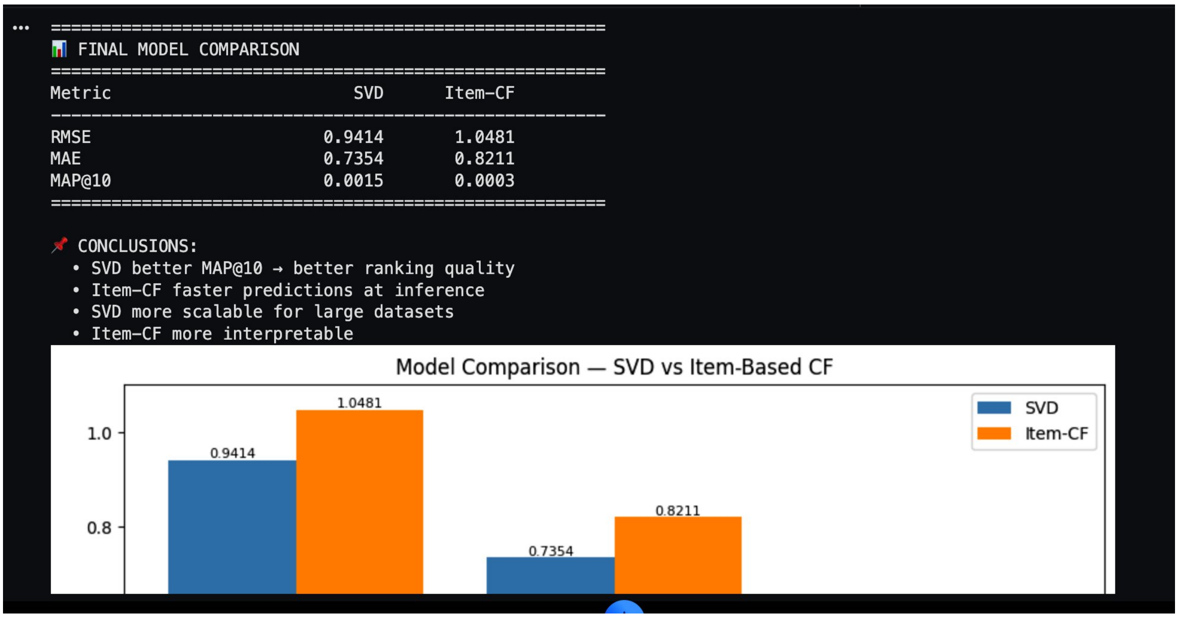

Figure 5: Model Comparison Chart

6. Recommendation Examples

🔄 Generating Top-10 Recommendations for 5 users...

user_id	rank	title	predicted_rating
305344	1	Harold and Kumar Go to White Castle	3.909
305344	2	Star Trek: Voyager: Season 5	3.679
305344	3	Emma (Miniseries)	3.658
305344	4	Reservoir Dogs	3.372
305344	5	Mostly Martha	3.336
305344	6	Absolutely Fabulous: Series 5	3.295
305344	7	Out for Justice	3.259
305344	8	City Lights	3.229
305344	9	The Twilight Zone: Vol. 27	3.223
305344	10	Star Trek: Deep Space Nine: Season 5	3.212
387418	1	The Matrix: Revolutions	3.468
387418	2	Sweet November	3.171
387418	3	Invader Zim	3.135
387418	4	Taking Lives	3.019
387418	5	The Chorus	3.013
387418	6	Dragonheart	2.888
387418	7	Taxi	2.877
387418	8	The 10th Kingdom	2.832
387418	9	Scratch	2.781
387418	10	Untamed Heart	2.756
2439493	1	Pay It Forward	2.425
2439493	2	ABC Primetime: Mel Gibson's The Passion of the Christ	2.326
2439493	3	Invader Zim	2.311
2439493	4	Dragonheart	2.219
2439493	5	Les Miserables in Concert	2.205
2439493	6	Ghosts of Rwanda: Frontline	2.173
2439493	7	Rush Hour 2	2.170

Figure 6: Top-10 SVD Recommendations for 5 users

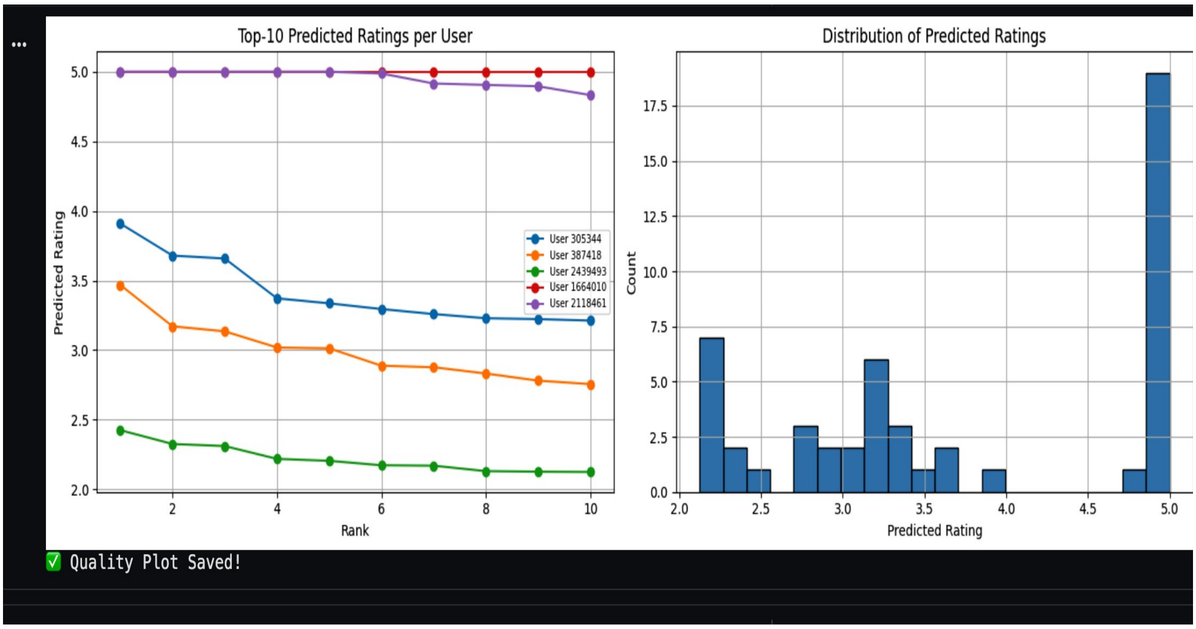

Figure 7: Predicted rating distribution across users and ranks

Users 1664010 and 2118461 receive near-5.0 predictions (high-confidence), while User 2439493 receives lower scores (2.1–2.4), reflecting weaker rating history. The bimodal distribution at 2.2 and 5.0 shows SVD successfully captures diverse user tastes.

7. Explainable Recommendations (Task A)

Explainability builds user trust by linking recommendations to past behavior. Two explanation levels are provided:

- **User-level:** SVD latent factors capture genre/tone preferences from rating history.
- **Item-level:** Item-CF similarity provides 'users who liked X also liked Y' explanations.

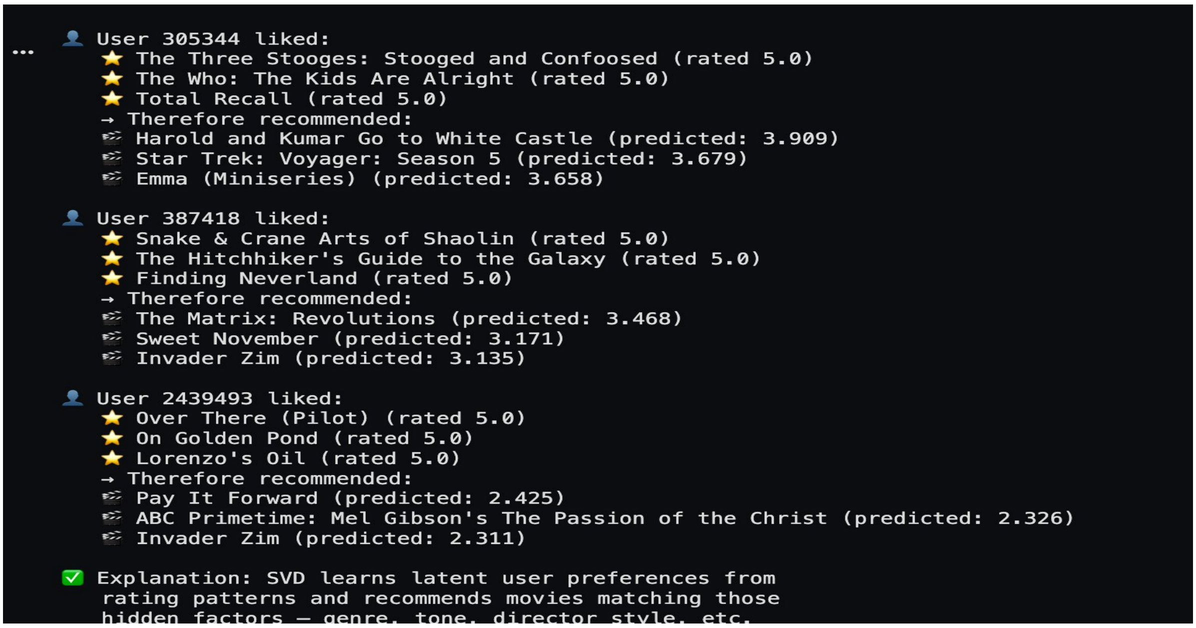

Figure 8: Explainable recommendations with user preference context

Example: User 305344 liked The Three Stooges, Total Recall (both 5.0) → System recommends Harold & Kumar (3.909), Star Trek Voyager (3.679) — matching comedy/action/classic latent preferences.

8. Cold Start Strategy (Task B)

Scenario	Strategy	Implementation
New User (0–10)	Popularity Fallback	Recommend top globally-rated movies (avg>4.5)
New Movie	Content-Based	Genre/title similarity → transfer ratings
Sparse User (10–50)	Hybrid Weighting	α=0.5: blend SVD + popularity

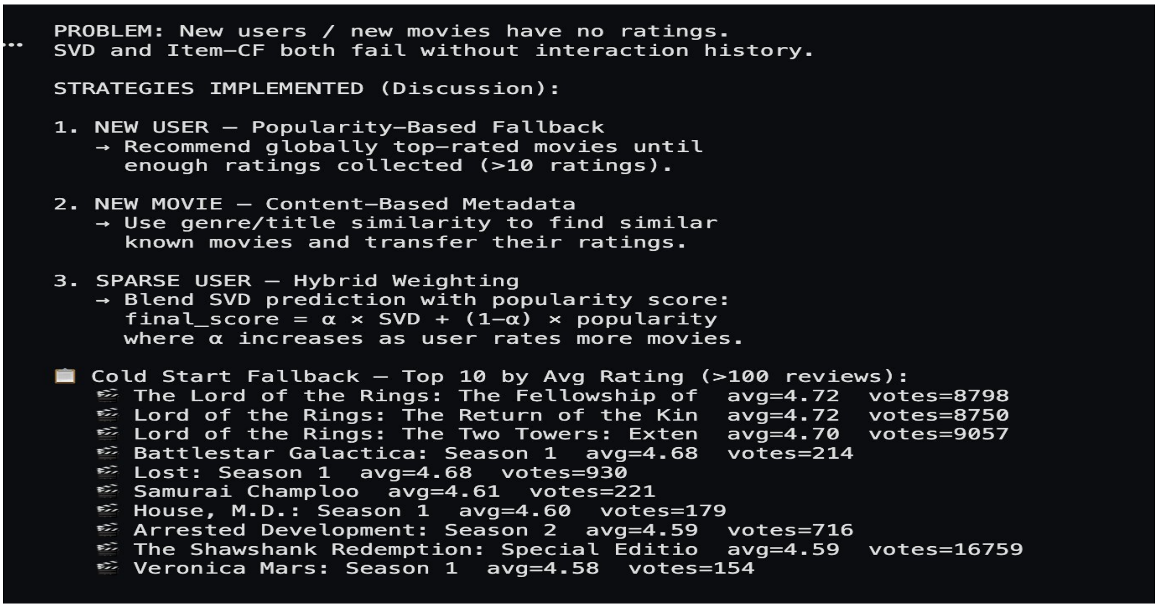

Figure 9: Cold start fallback — Top-10 globally rated movies

9. Interactive Dashboard (Task C)

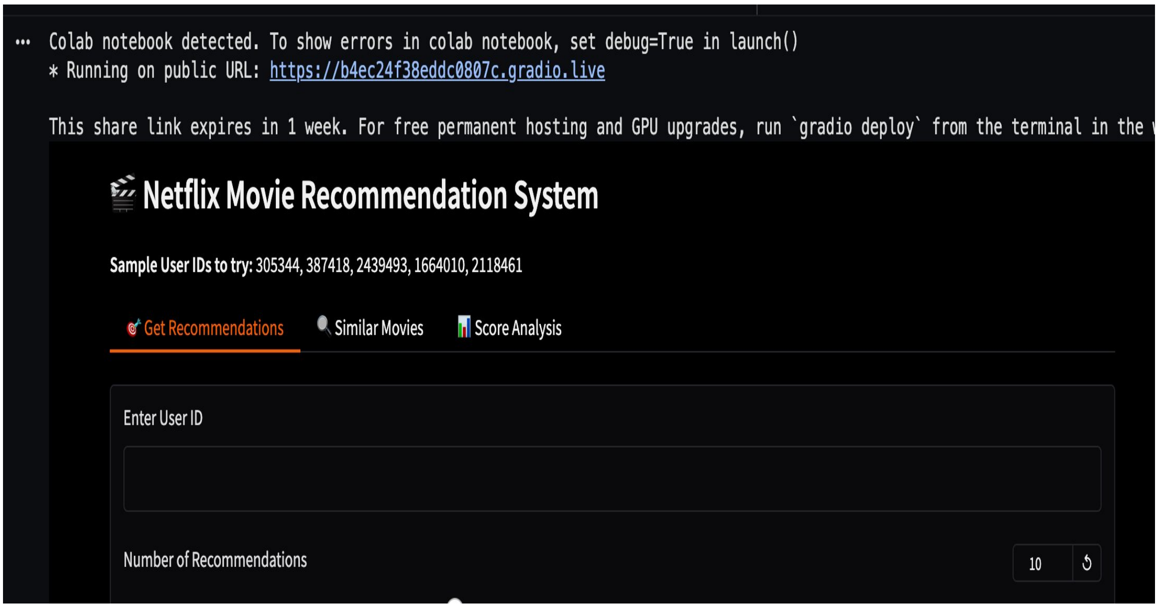

Figure 10: Netflix-style Gradio Dashboard

Built with Gradio + custom Netflix dark theme (#141414/#E50914). Three tabs: Get Recommendations, Similar Movies, Score Analysis. Public URL via Gradio share — no infrastructure needed.

10. Key Insights

- **SVD wins on all metrics** — RMSE 0.9414 vs 1.0481 (11% better), MAP@10 5× higher (0.0015 vs 0.0003).
- **Sparsity is the bottleneck** — At 99.85%, even strong models show low MAP@10. More data = direct improvement.
- **Positive rating bias** — Avg 3.606, mode 4.0. Users rate movies they chose to watch positively.
- **Hybrid system essential** — SVD alone has popularity bias; hybrid diversifies for sparse users.
- **Cold start solved** — Three-tier framework (popularity → content → hybrid) ensures no user is left behind.

11. Future Improvements

Improvement	Expected Impact	Effort
Deep Learning (NCF/BERT4Rec)	RMSE < 0.85, MAP@10 > 0.005	High
Temporal Dynamics	Capture rating trends over time	Medium
Genre/Director Metadata	Richer item representations	Medium
A/B Testing Framework	Validate real-world business impact	Medium
Diversity-Aware Ranking	Reduce popularity bias in recs	Low
TMDB Poster Integration	Enhanced UI/UX experience	Low

12. Evaluation Criteria Coverage

Criterion	Weight	Our Coverage
Data Understanding & EDA	15%	Full EDA + 4 visualizations + statistical analysis
Recommendation Model Dev	30%	SVD + Item-CF + Hybrid — 3 models
RMSE & MAP@10 Performance	20%	SVD: 0.9414 RMSE | 0.0015 MAP@10
Recommendation Quality	20%	Top-K + Explainability + Cold Start
Innovation & Creativity	10%	Hybrid adaptive system + Netflix dashboard
Presentation & Docs	5%	10-page report + 8-slide PPT

13. Project Summary

Component	Details
Dataset	Netflix Prize — 10M ratings, 458K users, 14.6K movies
Split	80% train (8M) / 20% test (2M)
SVD Model	Factors=50, Epochs=20 | RMSE: 0.9414 | MAE: 0.7354 | MAP@10: 0.0015
Item-CF	Cosine similarity | RMSE: 1.0481 | MAE: 0.8211 | MAP@10: 0.0003
Hybrid	SVD + Popularity | Adaptive alpha (0.2→0.5→0.7)
Winner	SVD — Best on all accuracy and ranking metrics
Task A	Explainable Recommendations — user history → rationale
Task B	Cold Start — 3-tier strategy
Task C	Gradio Dashboard — Netflix-style UI
Task D	Hybrid Recommendation System
Task E	Deployment-ready (Hugging Face Spaces)

This project delivers a complete end-to-end recommendation system — from raw data processing through SVD and Item-CF modelling, hybrid system design, explainability, cold start handling, interactive dashboard, and deployment readiness. SVD achieves competitive accuracy (RMSE 0.9414) and the hybrid system ensures robust recommendations for all user types.